

“Science is a field which **grows continuously** with **ever-expanding frontiers**. Further, it is truly international in scope.”

John Bardeen, 1972

turning it on or off. William Shockley, the scientist in charge, had designed a “field-effect” amplifier: he thought that creating a strong electrical field close to a cylinder coated in the semiconductor silicon would affect its conductivity. When the device did not work, Shockley asked Bardeen and another scientist on the team, Walter Brattain, to investigate.

In an intensive series of experiments, Bardeen, who handled the theoretical aspects of the research, and Brattain, who ran the laboratory, reworked the apparatus. They created a “point-contact” transistor, which used a contact made from thin gold foil to send a small current across the surface of a crystal of germanium (a semiconductor). This changed the semiconductor so that a larger current flowed out through a second gold contact, a hair’s breadth away from the first. They had, in 1947, created the first solid-state amplifier (without a vacuum tube). They called it a transistor (a contraction of “transfer and resistor”) because of its dual properties of electrical conductance and resistance.

Shockley was angry that he had not been part of the original invention and, working alone, developed an improved design; his much more commercially

successful “junction” transistor used a “sandwich” of different semiconductors. Initially adopted as amplifiers in hearing aids and radios, transistors soon found their way into computers, telephones, and almost every other electronic device, making Bardeen, Brattain, and Shockley’s work the basis for the 20th century’s electronics and computer revolutions.

Superconductors

Bardeen left Bell Laboratories in 1951 for a post as professor of electrical engineering at the University of Illinois. He returned to researching superconductors and why their conductivity increases at very low temperatures. He and two colleagues, Leon Cooper and John Robert Schrieffer, worked out what became known as the BCS theory (from their initials) of

